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Kinetic and Thermodynamic Study of Methylene Blue Adsorption onto Activated Carbon Obtained from the Peel of *musa paradisiaca*

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Abstract

In this work, we fabricated activated carbon using the peel of *musa paradisiaca* (banana) as the carbonaceous material source. The activated carbon was obtained after applied a carbonization process under nitrogen atmosphere at 723.15 K. The activated carbon was chemically modified using three chemical agents (citric acid, tartaric acid, and EDTA). The surface properties of the materials were characterized by nitrogen sorptometry at 77 K. Furthermore, we determined the zero-load point of all materials. The kinetic and isothermal behavior of the materials to remove methylene blue from aqueous solution was studied. The thermodynamic parameters of the process for all materials were determined by applying the van't Hoff equation. Results showed that after chemical activation, there was an increase in the content of oxygenated groups onto activated carbon. Furthermore, the BET surface area of activated carbon was reduced from 808 to 724 m² g⁻¹. The volume of micropores was smaller after chemical activation and the volume of mesopores was greater. The zero-load point of materials was in a range between 4.96 and 5.60. Kinetic and isothermal results showed that after chemical modification, the removal capacity increased from 30.2 for activated carbon to 52.6 mg g⁻¹ for activated carbon modified with EDTA. Finally, the thermodynamic parameters showed that methylene blue adsorption using all materials was an endothermic and spontaneous process; the ΔG° value of activated carbon was -4.35 kJ/mol, and the ΔG° value of activated carbon modified with EDTA was -6.28 kJ/mol.

Keywords: activated carbon; adsorption; isotherm; methylene blue



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1. Introduction

Water is an essential resource for human life, ecosystems, and global sustainable development. Its low availability on the planet, the rate of population growth, and poor waste management resulting from human activities have generated great concern around the world. There is an urgent need to mitigate the impact on this vital resource [1]. While industrial growth has contributed to the economic development of society, at the same time, the industry's activities have deteriorated water quality, generating different pollutants such as synthetic dyes. It is estimated that between 700,000 and 1,000,000 tons of these

compounds are produced annually for use in textile, rubber, paper, plastics, food, and cosmetics companies. After production processes, nearly 20% of dye production is discharged untreated or partially treated into bodies of water, contributing to the deterioration in water quality [2]. In the context of dye pollution, methylene blue (3,7-bis(dimethylamino) phenothiazine chloride, tetramethylthionine chloride), a cationic dye highly soluble and persistent in the medium, is used for dyeing different materials (e.g., wool, paper, silk, cotton). Concentrations ranging from 10 to 200 mg L⁻¹ in untreated wastewater have been found in the discharge of water [3,4]. This amount greatly exceeds the 1 mg/L concentration; a limit reported as toxic in aquatic ecosystems [5]. Different studies have shown that concentrations as low as 0.1 mg L⁻¹ of methylene blue (MB) can reduce the penetration of light into water by up to 60%, affecting aquatic photosynthesis [6]. At higher concentrations, it is toxic to fish, algae, and some bacteria beneficial to ecosystems [7]. Due to the risk of methylene blue on the environment, different technological strategies for its treatment have been investigated (e.g., coagulation–flocculation [8], reverse osmosis, advanced oxidation processes [9], degradation using fungi [10], bacteria, and algae, and adsorption [11,12]). Among options, the adsorption methodology has become one of the most researched depuration methodologies because of its versatility, low cost, high efficiency in removal implementation, and low risk to the environment. Among the large number of existing adsorbents (e.g., activated carbon [13], metal oxides [14], biochar [15], clays [16]), the activated carbon is a material of interest due to its versatility and ease of preparation, which allows the use of different materials as carbon sources, such as agricultural waste, derived from production chains [17,18]. Different authors have reported the advantages of using activated carbon as an adsorbent material for removal applications. Kundu et al. reported a maximum removal capacity of $q_e = 73.45 \text{ mg g}^{-1}$ to remove MB from water using activated carbon derived from the leaf of *Lantana camara* L. [19]. Huong et al. reported a maximum removal capacity of $q_e = 137 \text{ mg g}^{-1}$ to remove MB from water using activated carbon derived from Durian peel–seed [17]. Kataya et al. reported a maximum removal capacity of $q_e = 30.4 \text{ mg g}^{-1}$ to remove methylene blue from water using activated carbon derived from kitchen waste [20].

Between 2020 and 2022, Colombia was the second banana producer in Latin America, with an estimated production of 2.5 million tons. Once the pulp is consumed, the peel remains as waste, which is discarded, generating bio-accumulation. Most agriculture-dependent countries need to focus on the reuse of agricultural and agro-industry waste for a stronger economy. This amount of waste is an opportunity to add value to agro-industrial waste, promoting principles of the circular economy that contribute to the country's sustainable development [21]. In this work, we want to show the potential of banana peel (*musa paradisiaca*) waste as an activated carbon source for the removal of dyes from water.

In this work, the thermodynamic and kinetic study of the adsorption of MB on activated carbon obtained from banana peel (*musa paradisiaca*) was carried out to establish the potential of the banana peel as an activated carbon source.

2. Materials and Methods

2.1. Activated Carbon Synthesis

The banana peel samples were collected from the Bana's company, which produces patacones in Carrera 3A, Comuna 9-Picalaña, 740004 Ibagué, TOL, Colombia, latitude: 4.435253 | longitude: -75.196661. In the preparation of the activated carbon (AC), first, the peel was washed using bi-distilled water; next, the peel was cleaned, then dried. Then, the peel was grinded. After that, the particle size of the material was between 3 and 6 mm.

Finally, the sample was mixed with H_3PO_4 solution (32% *w/v*), in a ratio of (2 mL H_3PO_4 : 1 g material) at 358 K for 2 h.

In the second stage, first, the sample was carbonized in a carbolite horizontal furnace; the impregnated precursor was placed in a quartz vessel under a $\text{N}_{2(\text{g})}$ (99.9%) flow of 80 mL min^{-1} at a heating rate of $278.15 \text{ K min}^{-1}$ until reaching a temperature of 723.15 K. The sample was heated at this temperature for 2 h. Next, the activated carbon was washed with hot distilled water to remove residues until reaching a neutral pH and conductivity between 5 and $10 \mu\text{S cm}^{-1}$. Finally, the activated carbon was dried in an oven at 383.15 K for 12 h and stored in hermetically sealed plastic containers and a nitrogen atmosphere [22].

2.2. Chemical Modification of Activated Carbon

The activated carbon was chemically modified. For this purpose, three portions of activated carbon were put in contact, separately, with the chemical agents—citric acid (Merck $\geq 99\%$) 1 M (AC/CA), tartaric acid (Merck $\geq 99\%$) 1 M (AC/TA), and EDTA (Merck $\geq 99\%$) 1 M (AC/EDTA)—all in a ratio of 4 g of activated carbon in 25 mL of the corresponding solution, and each mixture was stirred for 30 min (Cole-Parmer 8852) and dried at 323.15 K for 12 h.

The modified AC was washed with deionized water until turbidity was not detected in the wash water (two drops of a 0.1 M lead nitrate solution were added to verify cleaning of wash water). Finally, modified AC was dried at 383.15 K for 2 h and stored in plastic containers under a nitrogen atmosphere [23–25].

2.3. Characterization of Activated Carbon

The textural characteristics of the materials were carried out by sorptometry with N_2 at 77.15 K. A prior degassing of the materials at 423.15 K was carried out by using the equipment Autosorb iQ2 (Quantachrome Instruments, New London, USA), apparent surface areas were calculated from the BET equation, and microporous volume V_o (N_2) and narrow microporosity volume V_n (CO_2) (pores $< 0.7 \text{ nm}$) were obtained by applying the Dubinin–Radushkevich equation to nitrogen adsorption data (Density of liquid $\text{N}_2 = 0.808 \text{ g cm}^{-3}$). The total pore volume (V_t) was calculated by the subtraction of volume adsorbed at a relative pressure of 0.99 ($V_{0.99}$) and the volume of mesopores. Furthermore, the chemical characteristics of porous solids were established by determining the zero-load point (P_{zc}) and Boehm titrations.

2.4. Isothermal, Kinetic, and Thermodynamic Study

In the kinetic study, MB (Merck $\geq 99\%$) (50 mg/L) was mixed with activated carbon (load = 100 mg, Volume = 25 mL, pH = 7.0, and temperature of 298 K). The MB concentration was determined every 10 min at 652 nm by spectrophotometry. The adsorption capacity of activated carbon was calculated following Equation (1):

$$q_t = \frac{((C_o - C_t)V)}{m}, \quad (1)$$

where C_o is the initial MB concentration and C_t is the MB concentration at every time. V is the volume of the solution and m is the activated carbon load. To model kinetic results, we applied three models: (i) the pseudo-first order (FO), (ii) the pseudo-second order (SO), and the intraparticle diffusion model. Details of mathematical equations are in the supporting information (Section S1).

In the isothermal study, we determined the adsorption capacity of MB on the activated carbon according to Equation (2):

$$q_e = \frac{((C_o - C_e)V)}{m}, \quad (2)$$

where q_e is the MB adsorbed on activated carbon (mg g^{-1}) at equilibrium and C_e is the MB concentration at equilibrium. To model isothermal results, we applied three models: (i) the Freundlich, (ii) the Langmuir, (iii) the Temkin isotherm models, and (iv) the Dubinin–Radushkevich models. Details of mathematical equations are in the supporting information (Section S1). We applied the average relative error (ARE) to determine the best-fitting isotherm. Finally, we calculated the thermodynamic parameters according to the van't Hoff equation (Equation (3)):

$$\ln K = \frac{\Delta S^o}{R} - \frac{\Delta H^o}{R} \left(\frac{1}{T} \right), \quad (3)$$

where T is the absolute temperature (K); R is the universal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$); and K is the thermodynamic equilibrium constant.

3. Results and Discussion

3.1. Textural Characterization

Figure 1 shows the nitrogen adsorption isotherms for the materials fabricated in this work. The activated carbon without chemical activation shows a type I isotherm according to the IUPAC classification, with a typical behavior of microporous solids whose structure has narrow porosity [26]. The adsorption isotherms of chemically modified activated carbon showed changes in the porous structure compared with activated carbon without modification. The modified materials present type IV isotherms, with an H4 hysteresis loop typical of micro–mesoporous materials [27].

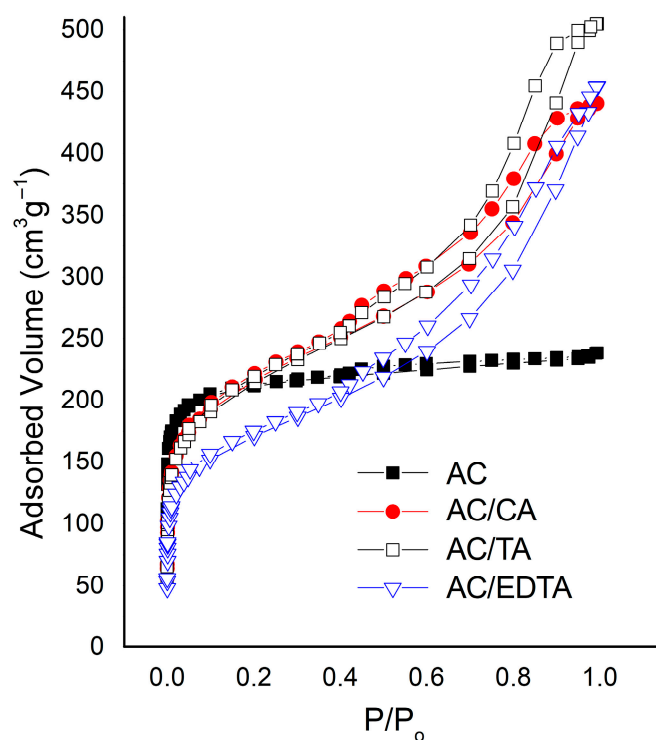


Figure 1. Nitrogen adsorption isotherms results for materials fabricated in this work.

Table 1 lists the textural characteristics of the materials. After chemical activation, the surface area of the materials decreased between 7.0 and 10% (the BET surface area range was reduced from 808 to 724 m² g^{−1}). Furthermore, the volume of micropores was smaller after chemical activation, and the volume of mesopores was greater. This trend has been reported previously after chemical activation [28,29]. Chemical modification introduces oxygenated functional groups into the carbon matrix, which may obstruct the porosity of the material. Furthermore, chemical activators can change the materials in two ways: (i) the chemical treatment can generate an obstruction by the size of the molecules, and (ii) chemical treatment can collapse the micropores, resulting in larger pores. This second possibility is confirmed by an increase in the volume of mesopores up to eight times compared with activated carbon without chemical modification [30,31].

Table 1. Results of nitrogen adsorption isotherms at 77 K.

Sample	S _{BET} (m ² g ^{−1}) *	V _O (cm ³ g ^{−1})	V _{meso} (cm ³ g ^{−1})	V _{0.99} (cm ³ g ^{−1})
AC	808	0.31	0.06	0.37
AC/CA	724	0.28	0.38	0.66
AC/EDTA	749	0.28	0.40	0.68
AC/TA	731	0.28	0.50	0.78

* data obtained from plotting Figure 1.

We calculated the pore size distributions (PSDs) of materials. For the determination of PSDs, two microscopic models were used to describe adsorption and fluid behavior in pores at the molecular level: Non-Local Density Functional Theory (NLDFE) and Quenched Solid Density Functional Theory (QSDFT) [31]. Figure 2 shows the PSDs of the materials using the QSDFT model, and Table 2 lists the results of the fitting.

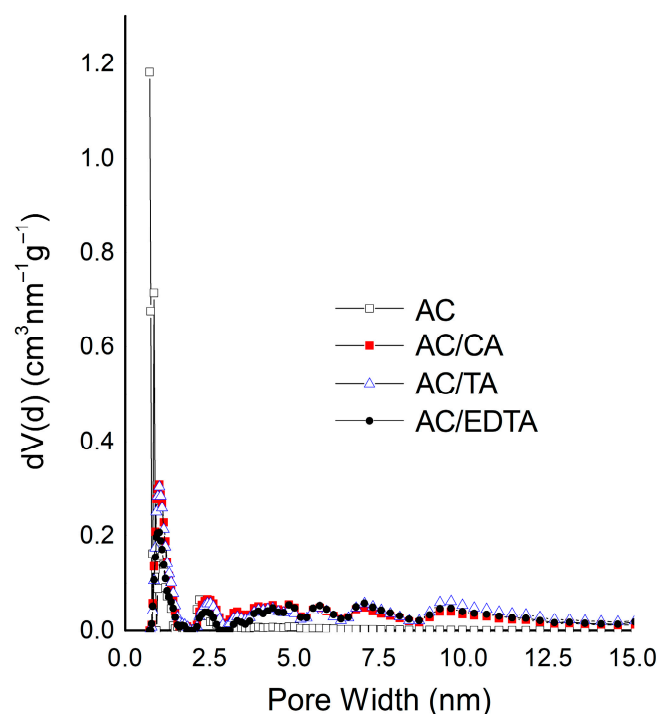


Figure 2. Pore size distribution (PSD) for fabricated materials.

Table 2. Chemical characteristics of activated carbons.

Sample	* BT ($\mu\text{mol g}^{-1}$)	** AT ($\mu\text{mol g}^{-1}$)	+ CG ($\mu\text{mol g}^{-1}$)	** LG ($\mu\text{mol g}^{-1}$)	++ PG ($\mu\text{mol g}^{-1}$)	P _{zc} (pH)
AC	70.65	241.53	48.05	102.76	90.72	5.60
AC/CA	116.6	363.29	70.52	195.68	97.09	5.15
AC/EDTA	102.4	389.25	94.70	185.74	108.81	4.96
AC/TA	91.40	300.61	65.60	129.31	105.70	5.38

* total basicity; ** total acidity. + carboxylic groups. ** lactonic groups. ++ phenolic groups.

In all samples, the QSDFT model suitably described the experimental data with an error rate between 0.31 and 0.78% compared to 1.25 and 2.45% calculated for the NLDFT model. The results showed a difference in pore size comparing materials after chemical modification with the starting material. The activated carbon had narrower porosity than the other materials, showing pore volumes below 2 nm. The chemically modified activated carbons showed pores of greater size > 20 nm, which is in accordance with the presence of mesopores in the carbonaceous structure. The chemical treatment applied to the starting material generated these changes in the textural parameters. Not only did the porosity of the surface change, but the chemistry of the surface also changed [31].

Table 2 lists the results of the chemical characteristics of materials (content of carboxylic groups, lactonic, phenolic, acidity, and total basicity). Results after the chemical activation process show an increase in the content of oxygenated groups. During chemical activation, the functional groups of the modifying agents can interact chemically with the active sites on the surface of carbon, generating an increase in the content of carboxylic, lactonic, and phenolic groups. Furthermore, the content of acid groups is consistent with the zero charge points obtained below a pH of 7 [32,33]. The AC/EDTA material presented the highest surface acidity and content of oxygenated groups, which may benefit the adsorption capacity of methylene blue dye of activated carbon, especially at pHs above the zero charge point, because the surface of the material will be negative, assisting electrostatic attractions with cationic species [34]. The same behavior applies to the other samples, whose surface chemistry is composed of a large number of oxygenated groups.

Figure 3 shows the FT-IR spectra of the original and modified activated carbons. The methods for modifying activated carbons with citric acid, tartaric acid, and EDTA used in the preparation of the activated carbons obtained in this investigation have been widely described in previous papers [35–37], which suggests the possible generation of chemical interactions between the activating agents and chemical groups present in the starting activated carbon surface. The FTIR spectra of the obtained activated carbons show the characteristic band of the stretching vibration of OH groups at 3205 cm^{-1} , which is slightly broader for AC/CA and AC/EDTA compared to AC, which would indicate an increase in carboxyl groups in these carbons as a product of chemical modification [38,39]. Additionally, bands found at 1693 cm^{-1} , usually due to the stretching vibration of the C=O bond in ketones, aldehydes, lactones, and carboxyl groups, and at 1565 cm^{-1} , attributed to the aromatic ring or the stretching vibration of the C=C bond, have higher relative intensities in AC/CA, AC/TA, and AC/EDTA carbons compared to AC, which suggests a significant increase in carbonyl. Furthermore, at 1150 cm^{-1} , it is possible to observe the characteristic band of the C–O tension, which is particularly more pronounced in the relative intensity for AC/CA, suggesting that this modification caused an increase in carboxylic groups [39,40]. These results concur with the results obtained from Boehm titration.

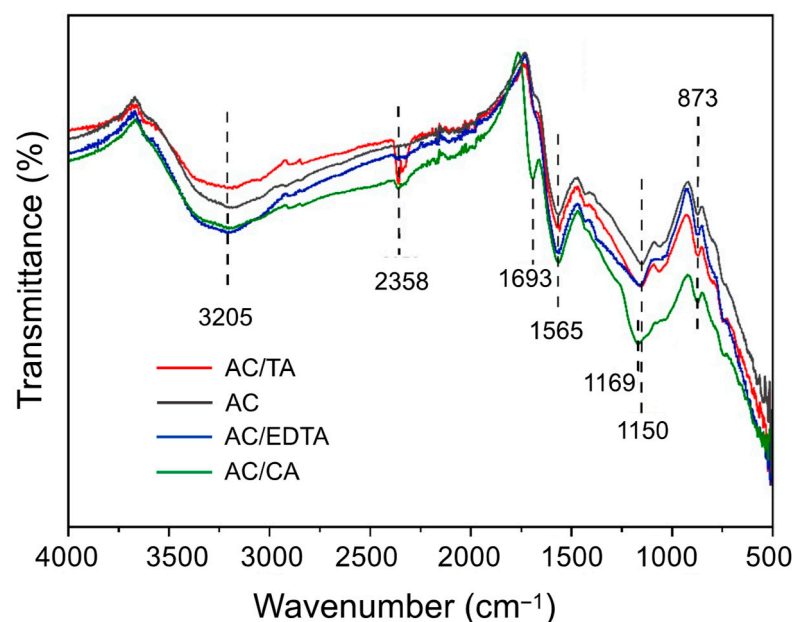


Figure 3. FT-IR spectra of original and modified activated carbons.

3.2. Kinetic Study

Figure 4 shows the experimental data and fitting results for MB adsorption on fabricated activated carbon, and Table 3 lists the fitting results. Among all models, the pseudo-first order (FO) was not suitable to describe the kinetic process. For all the chemical treatments, the pseudo-second order (SO) and intraparticle models showed the best-fitting results (see Table 3). The intraparticle model suggests that the diffusion within the pores of the activated carbon is an important step during the adsorption process. According to this model, if the adsorption process is only dependent on intraparticle diffusion, the y-axis intercept in the linear fitting will be zero [41–43]. A negative value of this constant (C value, intraparticle model, Table 3) indicates the presence of the effect of external film diffusion resistance [44,45]. In our case, only the AC/EDTA shows a positive C constant value (3.45 mg g^{-1}), suggesting that for this material, the film diffusion resistance is not a determining factor. However, for the other materials, the C constant values < 0 . It is known that the diffusion process is complex within pores of carbonaceous materials, and this process involves pore and film diffusion affecting the rate of the process, and it is possible that it did not follow the same pattern for different treatments [46]. According to Table 3, the SO model is also suitable to describe the adsorption process (high regression coefficient and low ARE value). In the SO model, during the adsorption process, the interaction is governed by the chemical interaction between MB molecules and active sites at the surface of the activated carbon [47]. Considering that the surface of activated carbon is heterogeneous and rough, it is possible that the MB adsorption follows a mixed kinetic where chemisorption is an important step but the adsorption rate is influenced by intraparticle diffusion. The electrostatic interaction between chemical groups located at the surface of activated carbon and MB molecules assists the adsorption, and in addition, the diffusion intraparticle of MB is also important during the process. The mixed regime, combining diffusion and the local adsorption reaction, has been previously reported; Castillo et al. reported a pseudo-heterogeneous model combining an apparent pseudo-second-order model with an intraparticle model for the removal of Rhodamine B on a zeolite [48,49]. Furthermore, Simic et al. reported multilinearity during adsorption of Pb onto Hydro-Pyrochar obtained from corn cob, indicating that more than one step was involved during the removal process [50]. In a mixed regime of adsorption, a possible diagram for methylene blue sorption onto activated carbon can involve the following: (I) the process begins with the external mass

transfer from the solution to the activated carbon surface. Different chemical groups assist the methylene blue in anchoring to the activated carbon surface. (II) The negative charge of the surface assists in electrostatic interactions, (III) π - π interactions, (IV) hydrogen bond, and (V) intraparticle diffusion into the activated carbon pores (see Figure S1).

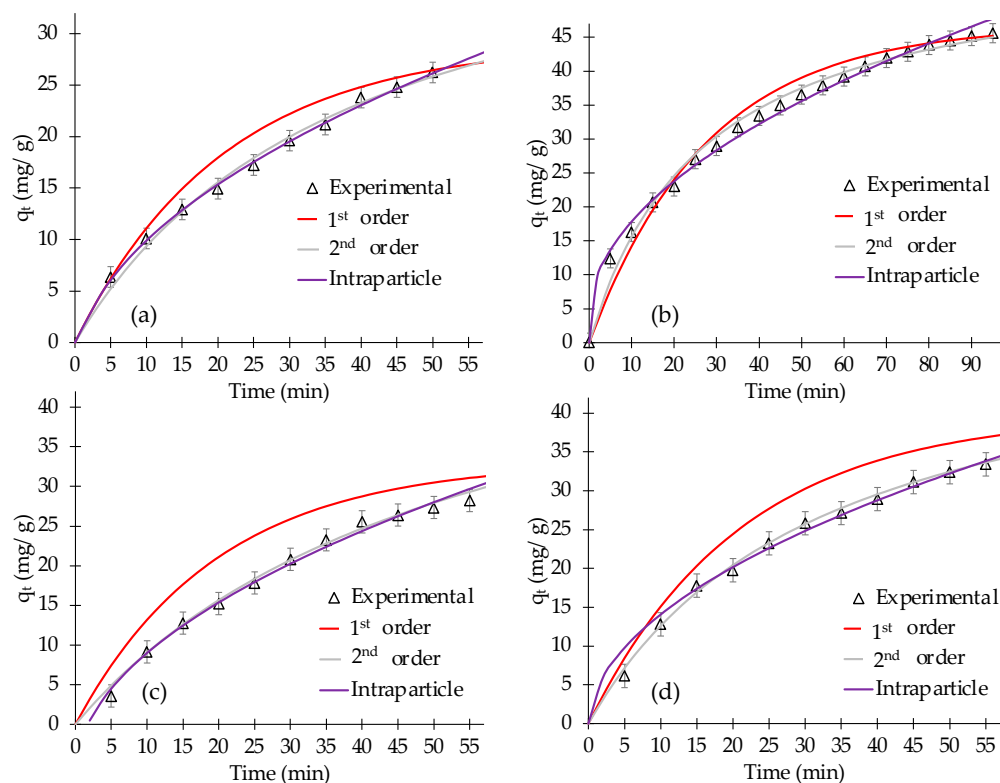


Figure 4. Kinetic fitting for experimental methylene blue adsorption onto (a) AC, (b) AC/EDTA, (c) AC/TA, and (d) AC/CA.

Table 3. Results of kinetic modeling for MB adsorption onto fabricated activated carbon.

Chemical Treatment	Model	Parameters *	R^2	ARE (%)
AC	FO	q_e (mg/g) 29.1 ± 2.9 k_1 (min^{-1}) 0.048 ± 0.005	0.956	10.8
	SO	q_e (mg/g) 46.1 ± 3.1 k_2 ($\text{g mg}^{-1} \text{min}^{-1}$) 0.0005 ± 0.0001	0.976	2.9
	Intraparticle	C (mg/g) -3.2 ± 0.8 k_{id} ($\text{mg g}^{-1} \text{min}^{-1/2}$) 4.15 ± 0.9	0.997	1.4
AC/EDTA	FO	q_e (mg g^{-1}) 46.7 ± 2.7 k_1 (min^{-1}) 0.036 ± 0.004	0.967	6.2
	SO	q_e (mg g^{-1}) 57.8 ± 4.4 k_2 ($\text{g mg}^{-1} \text{min}^{-1}$) 0.0006 ± 0.0001	0.987	3.7
	Intraparticle	C (mg/g) 3.4 ± 0.9 k_{id} ($\text{mg g}^{-1} \text{min}^{-1/2}$) 4.5 ± 0.8	0.989	2.9
AC/TA	FO	q_e (mg/g) 33.1 ± 2.5 k_1 (min^{-1}) 0.051 ± 0.007	0.979	25.9
	SO	q_e (mg/g) 58.5 ± 3.9 k_2 ($\text{g mg}^{-1} \text{min}^{-1}$) 0.0003 ± 0.0001	0.981	2.1
	Intraparticle	C (mg/g) -6.39 ± 1.1 k_{id} ($\text{mg g}^{-1} \text{min}^{-1/2}$) 4.86 ± 0.8	0.991	1.8

Table 3. Cont.

Chemical Treatment	Model	Parameters *		R ²	ARE (%)
AC/CA	FO	q _e (mg/g) 39.8 ± 2.7	k ₁ (min ^{−1}) 0.048 ± 0.007	0.984	16.7
	SO	q _e (mg/g) 53.5 ± 3.6	k ₂ (g mg ^{−1} min ^{−1}) 0.0006 ± 0.0001	0.995	1.7
	Intraparticle	C (mg/g) −0.6 ± 0.1	k _{id} (mg g ^{−1} min ^{−1/2}) 4.65 ± 0.7	0.991	2.7

* data obtained from plotting Figure 3.

3.3. Isothermal Study

Figure 5 shows the isothermal fitting results for all the fabricated materials. All of the isotherms showed an L-shape, suggesting that there is not a strong competition between water molecules and MB molecules for the active sites on the surface of activated carbon [51]. After chemical activation, the isothermal behavior was different for materials: (i) the isothermal behavior of AC and AC/CA was suitably described by the Freundlich model, and (ii) AC/EDTA and AC/TA was suitably described by the Langmuir model (see Table 4). These models rely on different assumptions: the Langmuir model occurs on a monolayer while the Freundlich model follows a multilayer adsorption process. However, comparing the K_L constants, the K_L is higher for EDTA and tartaric acid, suggesting that they have a higher affinity for MB compared with other materials. Furthermore, the chemical treatment with EDTA as a chemical activator of carbonous material showed the best result for the q_{max} value (52.6 mg g^{−1}). During the activation process, the carboxyl groups of EDTA can assist in the formation of functional groups with the O- and N-containing groups, improving the removal capacity of the adsorbent [52]. This result is in accordance with the kinetics results (AC/EDTA was the only material that did not show the effect of external film diffusion resistance). The q_{max} values obtained in this work were compared with the following previous reports: Kurnia et al. reported a q_{max} value of 79.3 mg g^{−1} for MB removal from water, using activated carbon obtained by physical activation of coconut shell, with a BET area of 467 m² g^{−1}, a pore volume of 0.27 cm³ g^{−1} and an oxygenated group content of 557 (μmol g^{−1}) [53]. Sawasdee et al. reported a q_{max} value of 26.31 mg g^{−1} for the removal of MB from water using activated carbon obtained by chemical activation with phosphoric acid from rice husks, with a BET area of 244 m² g^{−1}, a pore volume of 0.14 cm³ g^{−1}, and oxygenated and carboxylic, lactonic, and phenolic groups [54]. Dalmaz et al. reported a q_{max} value of 285 mg g^{−1} for the removal of MB from water using activated carbon prepared by chemical activation, with zinc chloride from used cigarette butts, obtaining a BET area of 667 m² g^{−1} and different oxygenated groups of hydroxyl-type lactonics [55]. It is observed that although the AC/EDTA carbon has a larger BET area than the carbons obtained in the studies by Dalmaz et al. and Kurnia et al., this material has a lower MB removal capacity. Therefore, it is possible to infer that the MB removal capacity is related not only to the surface area of the adsorbent materials but also to the surface chemistry, which influences the adsorption process. Furthermore, despite the activated carbons prepared in this study not exhibiting the highest MB adsorption capacity, they showed suitable removal performance, making them a viable option for removing the target pollutant.

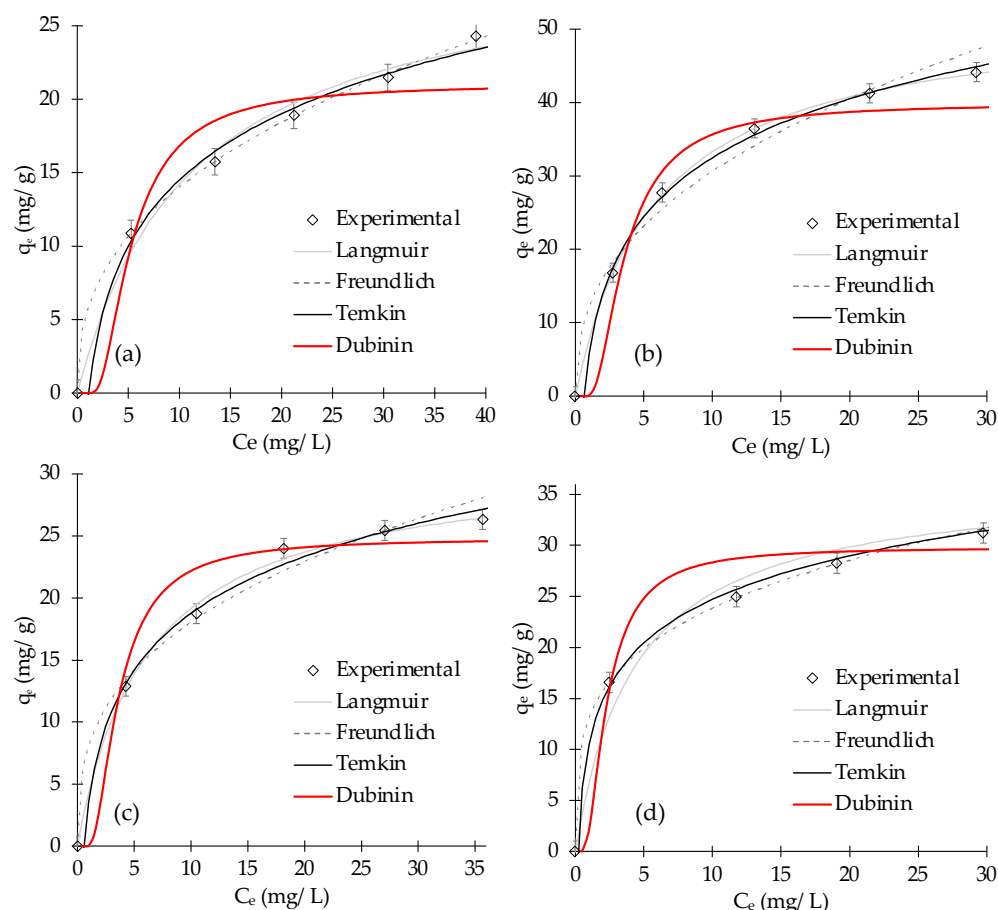


Figure 5. Isothermal fitting for experimental methylene blue adsorption onto (a) AC, (b) AC/EDTA, (c) AC/TA, and (d) AC/CA.

To study the recyclability of the modified activated carbon, three consecutive adsorption cycles were performed using the same sample (AC/EDTA). After the second MB adsorption cycle, the removal efficiency was reduced by 11%, and after the third cycle, the removal efficiency decreased by 43% relative to the initial run. This trend could be explained by two factors: (i) loss of the EDTA active sites on AC due to deterioration after washing between cycles; (ii) the aggregation or retention of MB molecules on the adsorbent surface, which prevents their removal by the washing stage prior to the next removal cycle. Similar behavior was reported by Alkhabbas et al., who reported an adsorption decreasing from 99.4% to 58% after five consecutive adsorption cycles of the naphthol blue black on activated carbon obtained from oak cupules [56].

Table 4. Results of isotherm modeling for MB adsorption onto fabricated activated carbon.

Material	Model	Parameters *			
AC	Langmuir	$q_{\max}$ (mg/g)	k_L (L/min)	R^2	ARE (%)
		30.2 ± 2.8	0.09 ± 0.01	0.984	4.5
	Freundlich	K_F (mg/g) $(L/g)^{1/n}$	$1/n_f$		
		5.6 ± 0.5	0.40 ± 0.02	0.999	0.5
	Temkin	A_T (L/g)	B_T (J/mol)		
		5.5 ± 0.5	6.2 ± 0.9	0.990	2.9
	Dubinin	q_s (mg/g)	K_D (mol ² kJ ²) $\times 10^{-6}$		
		21.0 ± 1.4	4	0.843	8.5

Table 4. Cont.

Material	Model	Parameters *			
AC/ EDTA	Langmuir	$q_{\max}$ (mg/g) 52.6 ± 2.3	k_L (L/min) 0.17 ± 0.02	0.999	0.4
	Freundlich	K_F (mg/g) $(L/g)^{1/n_f}$ 12.1 ± 1.1	$1/n_f$ 0.40 ± 0.02	0.964	5.1
	Temkin	A_T (L/g) 1.7 ± 0.3	B_T (J/mol) 11.6	0.996	1.7
	Dubinin	q_s (mg/g) 39.8 ± 3.2	K_D (mol ² kJ ²) $\times 10^{-6}$ 2.0 ± 0.1	0.927	9.6
AC/TA	Langmuir	$q_{\max}$ (mg g ⁻¹) 31.1 ± 1.6	k_L (L min ⁻¹) 0.16 ± 0.02	0.998	1.8
	Freundlich	K_F (mg g ⁻¹) $(L g^{-1})^{1/n_f}$ 8.1 ± 0.9	$1/n_f$ 0.35 ± 0.02	0.966	3.4
	Temkin	A_T (L g ⁻¹) 1.7 ± 0.3	B_T (J mol ⁻¹) 6.6 ± 1.0	0.980	2.1
	Dubinin	q_s (mg/g) 24.8 ± 2.4	K_D (mol ² kJ ²) $\times 10^{-6}$ 2 ± 0.1	0.912	7.0
AC/CA	Langmuir	$q_{\max}$ (mg g ⁻¹) 36.4 ± 1.9	k_L (L min ⁻¹) 0.23	0.992	6.0
	Freundlich	K_F (mg g ⁻¹) $(L g^{-1})^{1/n_f}$ 13.1 ± 1.0	$1/n_f$ 0.26 ± 0.03	0.999	0.5
	Temkin	A_T (L g ⁻¹) 5.5 ± 0.4	B_T (J mol ⁻¹) 6.15 ± 1.1	0.980	1.8
	Dubinin	q_s (mg/g) 29.8 ± 2.3	K_D (mol ² kJ ²) $\times 10^{-6}$ 0.9 ± 0.1	0.874	6.6

* data obtained from plotting Figure 4.

3.4. Thermodynamic Study

Figure S2 shows the plot of linear fitting to the Arrhenius equation, and Table 5 lists the thermodynamic results. The thermodynamic results indicated that adsorption of MB onto all materials was spontaneous ($\Delta G < 0$) independently of the chemical agent employed for activation (see Table 5). Results showed that after increasing the temperature of the process, the spontaneity of the blue methylene adsorption onto materials increases. Furthermore, the material activated with EDTA was the material with the lowest ΔG value (see Table 5), indicating that the MB adsorption onto this material was the most spontaneous process among all materials studied in this work, and in addition, this material reported the highest $q_{\max}$ among all fabricated materials. The positive value for entropy can be associated with the liberation of water hydration molecules into the solution from MB and the active sites of activated carbon. Finally, the MB adsorption onto all of the materials was endothermic in nature. Table 5 lists the thermodynamic adsorption values for MB adsorption onto modified activated carbon reported by other authors. Results presented in this work are comparable with those reported previously.

Table 5. Thermodynamic adsorption values for MB adsorption onto modified activated carbon.

Adsorbent Materials	Temperature (K)	Thermodynamic Parameters		
		ΔG (kJ mol ^{−1})	ΔH (kJ mol ^{−1})	ΔS (J mol ^{−1} K ^{−1})
AC (This work)	303	−4.35	29.4	110.0
	313	−5.48		
	323	−6.97		
	333	−7.55		
AC/EDTA (This work)	303	−6.28	23.38	100.0
	313	−7.22		
	323	−7.98		
	333	−9.30		
AC/TA (This work)	303	−5.27	27.70	110.0
	313	−5.87		
	323	−7.17		
	333	−8.47		
AC/CA (This work)	303	−6.48	17.60	80.0
	313	−6.95		
	323	−7.88		
	333	−8.82		
Activated carbon [57]	293	−9.75	51.2	−143
	303	−6.89		
	313	−5.74		
	323	−5.45		
Activated carbon/CS * [53]	303	2.62	27.99	80.4
	313	1.80		
	323	1.23		
	328	0.48		
** LC-Biochar [58]	303	−11.67	17.59	100
	313	−13.19		
	323	−13.62		

* CS: coconut shell. ** LC: lignocellulosic.

4. Conclusions

In this study, activated carbon was obtained from the peel of *musa paradisiaca* and evaluated for its potential in methylene blue removal from aqueous solutions. Textural characterization revealed that the topology significantly changed after chemical modification of the activated carbon. After chemical activation, the surface area of materials decreased between 7.0 and 10%. Furthermore, the volume of mesopores increases after chemical activation from 0.06 to 0.50 cm³ g^{−1}. Chemical characterization showed that the AC/EDTA material presented the highest surface acidity and content of oxygenated groups, which may favor the adsorption capacity of methylene blue dye from water. The kinetic and isothermal study showed that AC/EDTA had the best performance in the removal of methylene blue from water. Results suggest that the chemically modified activated carbon obtained from the peel of *musa paradisiaca* was an effective sorbent for methylene blue removal from water, showing a removal capacity of 52.6 mg/g. This result is comparable with previous reports and demonstrates the potential of this local agricultural waste to become a source of carbonaceous sorbent for developing such applications.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/sci7040170/s1>, Section S1: Kinetic and isothermal adsorption models [59–63]; Figure S1: Possible diagram for methylene blue sorption onto activated carbon. Figure S2: Fitting of van't Hoff equation for adsorption of MB onto fabricated materials.

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